17/12/2025

Code-A Phase-1

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 NCERT Booster Program for NEET-2026_NBTS-02A **Time : 180 Min.**

Topics Covered:
Physics: Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles and rotational motion
Chemistry: Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics
Botany: The Living World, Biological Classification, Morphology of Flowering Plants
Zoology: Biomolecules, Breathing and Exchange of Gases, Body Fluids and Circulation, Excretory Products and Elimination

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet.and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A body is thrown horizontally with speed u from the top of a tower and it strikes the ground after four second at an angle of 53° with the horizontal. The speed of projection of the body u is

   (1) 40 m s^{-1}

   (2) 30 m s^{-1}

   (3) 50 m s^{-1}

   (4) 25 m s^{-1}

2. Rain is falling vertically with 4 km h^{-1} (–y direction). If a man starts walking with 7 km h^{-1} in +x direction, then velocity of rain as seen by man will be

   (1) $-\left(7\hat{i} + 4\hat{j}\right)$ km h^{-1}

   (2) $\left(7\hat{i} + 4\hat{j}\right)$ km h^{-1}

   (3) $\left(-7\hat{i} + 4\hat{j}\right)$ km h^{-1}

   (4) $\left(7\hat{i} - 4\hat{j}\right)$ km h^{-1}

3. This is a match the column type question, which has 2 Columns (Column I and Column II). Column I has four entries (A), (B), (C) and (D), Column II has four entries (P), (Q), (R) and (S). Match the entries in Column I with the entries in Column II. Each entry in Column I may match with one or more entries in Column II. Four choices are given for this question, out of which ONLY ONE is correct.
Match the columns and choose the correct option.

	Column -I		Column -II
(A)	A particle is moving in a straight line	(P)	The acceleration vector may be in the direction of velocity vector
(B)	A particle is projected from ground at an angle $\theta\left(0<\theta<\frac{\pi}{2}\right)$	(Q)	The acceleration vector is always perpendicular to its velocity vector
(C)	A particle is performing uniform circular motion	(R)	The acceleration vector may be opposite to velocity vector
(D)	A particle is performing non uniform circular motion	(S)	The acceleration vector may be at some angle θ with the velocity vector $\left(0<\theta<\frac{\pi}{2}\right)$

(1) A(P, R); B(S); C(Q); D(S)

(2) A(P); B(Q); C(Q); D(S)

(3) A(R); B(Q); C(S); D(Q)

(4) A(P, Q); B(R, S); C(Q); D(Q)

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4. A particle is moving in xy-plane in a circular path with centre at origin. If at an instant the position of particle is given by $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$, then velocity of particle is along

(1) $\frac{1}{\sqrt{2}}\left(\hat{i}-\hat{j}\right)$

(2) $\frac{1}{\sqrt{2}}\left(\hat{j}-\hat{i}\right)$

(3) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$

(4) Either (1) or (2)

5. The equation of a ground-to-ground projectile is given as $y=\sqrt{3}x-\frac{gx^2}{2}$.The velocity of projection of the projectile is (where symbols have their usual meaning)

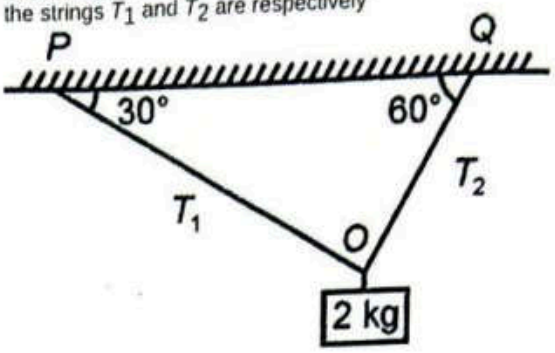
(1) 4 m/s

(2) 8 m/s

(3) 2 m/s

(4) 1 m/s

6. A block of mass 2 kg hangs in equilibrium with the help of two strings *OP* and *OQ* as shown in figure. The tension in the strings T_1 and T_2 are respectively



(1) 10 N, 10 N

(2) 10 N, $10\sqrt{3}$ N

(3) $10\sqrt{3}$ N, $10\sqrt{3}$ N

(4) 20 N, 20 N

7. Consider the following statements out of which one is labelled as Assertion (A) and other as reason (R).
Assertion (A) : Two identical masses having same speed may have different linear momentum.
Reason (R) : Linear momentum is a vector quantity and inspite of same magnitude, different direction renders the momenta unequal.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

8. A bullet is fired with a speed 200 m/s. If the mass of bullet is 100 gm and of gun is 2 kg, then the recoil speed of gun is

(1) 20 m/s

(2) 2 m/s

(3) 10 m/s

(4) 5 m/s

9. A machine gun has a mass 5 kg. It fires bullets of mass 50 g each at the rate of 30 bullets/minute with a speed of 400 m/s. Force required to hold the gun in position is

(1) 150 N

(2) 15 N

(3) 10 N

(4) 20 N

10. An unbanked curve has a radius of 40 m. The maximum speed at which a car can make a turn if the coefficient of static friction is 0.50 is nearly

(1) 7 m/s

(2) 14 m/s

(3) 21 m/s

(4) 28 m/s

11. Three forces $\vec{F}_1, \vec{F}_2, \vec{F}_3$ are acting on a particle of mass m such that $\vec{F}_2$ and $\vec{F}_3$ are mutually perpendicular and the particle is in equilibrium under these three forces. If the force $\vec{F}_1$ is removed, then the magnitude of acceleration of the particle is

(1) $\frac{F_1+F_2}{m}$

(2) $\frac{F_2+F_3}{m}$

(3) $\frac{F_2+F_3-F_1}{m}$

(4) $\frac{\sqrt{F_2^2+F_3^2}}{m}$

12. The coefficient of restitution 'e' for a perfectly inelastic collision is

(1) Zero

(2) 1

(3) $\frac{1}{2}$

(4) ∞

13. Two balls each of mass m are moving towards each other with same speed v. If the collision between them is perfectly inelastic, then the loss in the kinetic energy due to collision is

(1) $\frac{1}{2}mv^2$

(2) mv^2

(3) $2mv^2$

(4) $\frac{mv^2}{4}$

14. A small ball of mass 10 kg is moving with speed 4 m/s and collides elastically with another ball of mass 6 kg at rest. If the collision is head-on, then the respective speeds of the balls after the collision will be

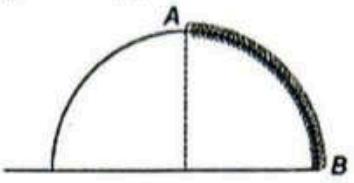
(1) 1 m/s, 5 m/s

(2) 6 m/s, 4 m/s

(3) 2 m/s, 6 m/s

(4) 2 m/s, 1 m/s

15. A uniform chain AB of mass m and length l is placed with one end A at the highest point of hemisphere of radius R as shown below. If the bottom of hemisphere is taken as zero potential energy level, then potential energy of chain is $\left(\text{Given } l = \frac{\pi R}{4}\right)$



(1) mgR

(2) $\frac{2mgR}{\pi}$

(3) $\frac{mgR}{2\pi}$

(4) $\frac{4mgR}{\pi}$

16. A force $\vec{F} = (\hat{i}+\hat{j}+2\hat{k})$N is acting on a particle moving with constant velocity $\vec{v} = (4\hat{i}+\hat{j}-\hat{k})$m/s. The power delivered by the force is

(1) 6 W

(2) 1 W

(3) 4 W

(4) 3 W

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17. If keeping moment of inertia constant, the angular momentum of a body is increased by 20%. The percentage increase in its rotational kinetic energy is

(1) 40%

(2) 12%

(3) 44%

(4) 20%

18. A constant torque of 100 N m is acting on a rigid body of moment of inertia 20 kg m^2 about an axis passing through its centre of mass. Its angular velocity after 4 s starting from rest will be

(1) 10 rad/s

(2) 20 rad/s

(3) 40 rad/s

(4) Zero

19. A force $\left(2\hat{i}-4\hat{j}-\hat{k}\right)$ N acts on a particle having position vector $\left(2\hat{i}+4\hat{k}\right)$ m. The torque acting on the particle about origin will be

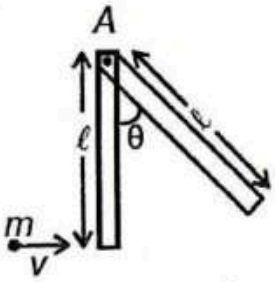
(1) $\left(-16\hat{i}-6\hat{j}+8\hat{k}\right)$ N m

(2) $\left(16\hat{i}+10\hat{j}-8\hat{k}\right)$ N m

(3) $\left(-6\hat{i}-16\hat{j}+8\hat{k}\right)$ N m

(4) $\left(16\hat{i}+6\hat{j}+8\hat{k}\right)$ N m

20. A rod of mass m is hinged at upper end A as shown in the figure. A point object having mass m strikes at lower end and stops. The maximum angle through which rod will rotate is $\left(\text{Given } v^2 < \frac{g\ell}{3}\right)$



(1) $\cos^{-1}\left(1-\frac{3v^2}{2g\ell}\right)$

(2) $\cos^{-1}\left(1-\frac{3v^2}{g\ell}\right)$

(3) $\cos^{-1}\left(1-\frac{v^2}{2g\ell}\right)$

(4) $\cos^{-1}\left(1-\frac{v^2}{g\ell}\right)$

21. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark. Find the value of 'm' such that the rod is in equilibrium. (g = 10 m/s^2)

(1) $\frac{1}{12}$kg

(2) $\frac{1}{2}$kg

(3) $\frac{1}{3}$kg

(4) $\frac{1}{6}$kg

22. **Statement (A) :** If a vector is rotated through an angle other than multiple of 2π, it changes.

Statement (B) : Two vector $n\vec{A}$ and $\vec{A}$ have opposite directions if n is a negative number.

Statement (C) : Two vectors of magnitude 4 N and 6 N are inclined at an angle of 60° with each other, then angle between their resultant vector and the 6 N vector is $\tan^{-1}\left(\frac{3\sqrt{3}}{7}\right)$

Consider the given statements and choose the correct option.

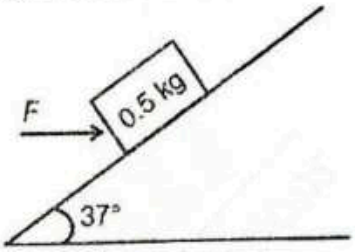
(1) Only statement (A) is correct

(2) All the statements are correct

(3) Only statement (C) is correct

(4) Only statement (A) and statement (B) are correct

23. A block of mass 0.5 kg is kept stationary on a smooth inclined plane by applying a horizontal force F as shown in figure. The value of F is



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(1) $\frac{15}{4}$ N

(2) 4 N

(3) $\frac{13}{3}$ N

(4) $\frac{16}{5}$ N

24. A particle is projected from the top of a cliff of height 60 m with speed u at an angle of 45° above the horizontal. It strikes the horizontal ground at a point 30 m horizontally from the base of the cliff. The magnitude of initial speed u is

(1) 8 m/s

(2) 10 m/s

(3) 12 m/s

(4) 14 m/s

25. A particle is projected from ground level with initial speed u at angle θ above horizontal. A vertical wall of height 16 m stands at horizontal distance 8 m from the launch point. The projectile just grazes the top of the wall and then lands on level ground at a horizontal distance 24 m from the launch point. The value of initial speed u is (neglect air resistance).

(1) 16 m/s

(2) 22 m/s

(3) 20 m/s

(4) 18 m/s

26. A modern grand-prix car of mass m = 800 kg travels on a flat track around a circular arc of radius R = 50 m at a speed v = 72 km/h. Aerodynamic devices produce a negative lift F_L (i.e., a downward aerodynamic force), increasing the normal load on tyres. The coefficient of static friction between tyres and track is μ_S = 0.8. Assume forces on all four tyres are identical and frictional force provides the required centripetal force. The magnitude of negative lift F_L (in newton) is (g = 9.8 m/s^2)

(1) 160 N

(2) 320 N

(3) 640 N

(4) Zero

27. **Statement (A) :** If resultant of two unit vectors is another unit vector, then angle between them, $\theta = 120°$

Statement (B) : If $\frac{|\vec{a}+\vec{b}|}{|\vec{a}-\vec{b}|}=1$, then the angle between $\vec{a}$ and $\vec{b}$ is zero $\left(|\vec{a}|, |\vec{b}| \neq 0\right)$

Consider the given statements and choose the correct option.

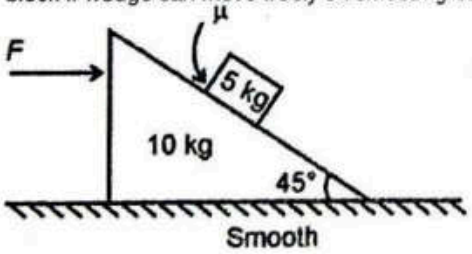
(1) Statement (A) is false and statement (B) is true

(2) Both statements are true

(3) Both statements are false

(4) Statement (A) is true and statement (B) is false

28. The surface between wedge and block as shown below, is rough (coefficient of friction μ = 0.5). The maximum force F for which there is no relative motion between wedge and block if wedge can move freely on smooth ground, is



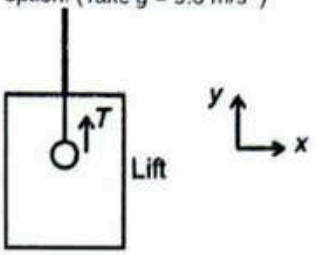
(1) 450 N

(2) 300 N

(3) 150 N

(4) 600 N

29. A pendulum bob of mass 50 gm is hanging from the ceiling of a lift by a light, inextensible string. Choose the incorrect option. (Take g = 9.8 m/s^2)



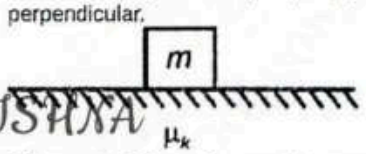
(1) When the lift has an acceleration of $(1.2\ m/s^2)\hat{j}$, the tension in the string is 0.55 N.

(2) When the lift has a constant velocity $(1.2\ m/s)\hat{j}$, the tension in the string is 0.49 N.

(3) When the lift has an acceleration $(-1.2\ m/s^2)\hat{j}$, the tension in the string is 0.43 N

(4) When the lift has a constant velocity $(-1.2\ m/s)\hat{j}$, the tension in the string is 0.55 N.

30. **Statement (A) :** For $f_k = \mu_k N$, $\vec{f}_k$ and $\vec{N}$ are always perpendicular.



Statement (B) : $f_k = \mu_k N$ is not a vector equation, it is scalar relationship between the magnitudes of the two forces.

Consider given statement and choose the correct option.

(1) Statement (A) is true but statement (B) is false

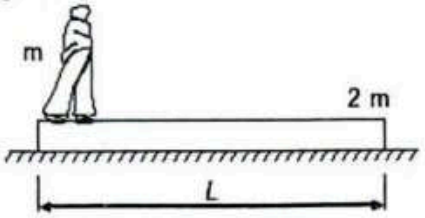
(2) Statement (A) is false but statement (B) is true

(3) Both statements are false

(4) Both statements are true

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31. A plank of mass $2m$ and length L is at rest on a frictionless horizontal floor. At one end of it, a child of mass m is standing as shown in figure. If child walks and reaches the other end, then the distance travelled by the child w.r.t ground will be



(1) $\frac{L}{3}$

(2) $\frac{L}{2}$

(3) $\frac{2L}{3}$

(4) $\frac{3L}{4}$

32. A block B of mass 2 kg moves rightwards with speed 15 m/s and collides elastically with a block A of mass 4 kg initially at rest on a horizontal rough floor. After collision, B reverses direction and moves leftward with speed 5 m/s, (coefficient of kinetic friction, $\mu_k = 0.2$). The distance s travelled by A (from the instant just after collision) before it comes to rest is

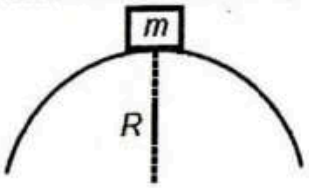
(1) 15 m

(2) 20 m

(3) 25 m

(4) 50 m

33. A block kept on a vertical hemisphere of radius R starts sliding down from rest. The angle with vertical at which it looses the contact with hemisphere is



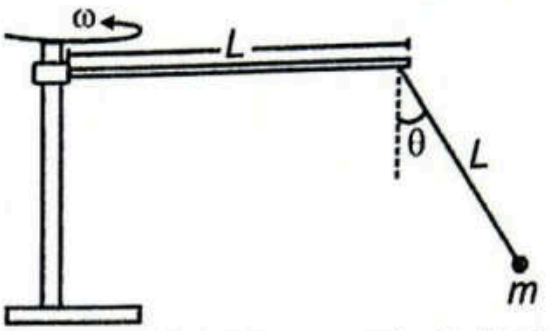
(1) $\theta = 60°$

(2) $\theta = \frac{\pi}{2}$

(3) $\theta = \sin^{-1}\left(\frac{2}{3}\right)$

(4) $\theta = \cos^{-1}\left(\frac{2}{3}\right)$

34.



A rod of length L is pivoted near one end and rotates in a horizontal plane with a constant angular speed ω. From free end of rod, a ball of mass m is suspended by light string of length L. In steady motion $\theta = 45°$, then ω(rad/s) is (take L = 25 cm)

(1) $\sqrt{\frac{2g\sqrt{2}}{1+\sqrt{2}}}$

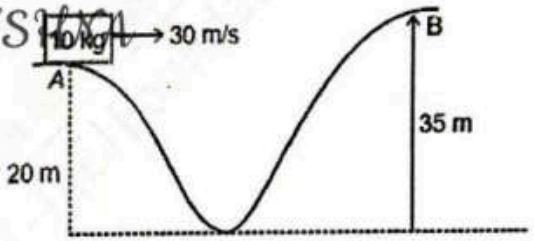
(2) $\sqrt{\frac{4g\sqrt{2}}{1+\sqrt{2}}}$

(3) $\sqrt{\frac{2g\sqrt{2}}{\sqrt{2}-1}}$

(4) $\sqrt{\frac{4g\sqrt{2}}{\sqrt{2}-1}}$

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35.



A block is given a velocity of 30 m/s on a smooth track, as shown. The velocity when it reaches point B is (g = 10 m/s^2)

(1) $\sqrt{200}$ m/s

(2) $\sqrt{600}$ m/s

(3) $\sqrt{40}$ m/s

(4) Zero

36. Consider the given statements and choose the correct option.

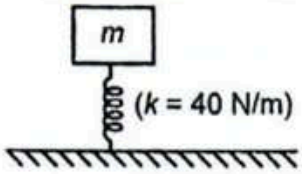
(1) All forces are conservative

(2) Work done against a non-conservative force is dissipated as heat energy

(3) Potential energy increases, if force field is attractive in nature

(4) Work done in a round trip is zero, in non-conservative force field

37. When a block of mass m is placed upon a spring (initially at natural length) of force constant k, the block compresses the spring by 0.5 m and comes to rest. Then m will be



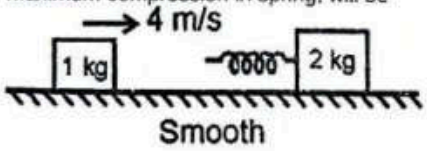
(1) 4 kg

(2) 1 kg

(3) 5 kg

(4) 6 kg

38. A block of mass 2 kg is resting on a smooth horizontal surface. A massless spring of stiffness k = 3 N/m is attached to the block as shown. Another block of mass 1 kg is moving with a speed 4 m/s towards the stationary block. The maximum compression in spring, will be



(1) $\frac{4}{3}\sqrt{2}$ m

(2) $\frac{\sqrt{2}}{3}$ m

(3) $\frac{\sqrt{3}}{2}$ m

(4) $\frac{\sqrt{3}}{4}$ m

39. A rigid body is rotating with a constant angular speed ω about a fixed vertical axis (taken as z-axis). The body is made up of many particles and for every particle of mass m at a distance r from axis moving with velocity $\vec{v}$, there exists another particle at diametrically opposite point moving with velocity $-\vec{v}$. If $L_\perp$ is component of total angular momentum perpendicular to axis of rotation and L_Z is the component along axis, then (I is MOI of the body about Z axis)

(1) $L_\perp \neq 0$ and $L_Z = 0$

(2) $L_\perp = 0$ and $L_Z = I\omega$

(3) Both $L_\perp$ and L_Z are zero

(4) $L_\perp \neq 0$ and $L_Z = \frac{I\omega}{2}$

40. Four particles of masses 2 kg, 4 kg, 6 kg and 8 kg are placed on a straight line at positions x = 1 m, x = 3 m, x = 7 m and x = 9 m respectively. A single particle of mass M is to be placed at x = 5 m such that centre of mass of all the five particles together is exactly at x = 6 m. The value of M is

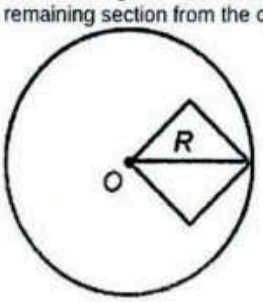
(1) 2 kg

(2) 6 kg

(3) 8 kg

(4) 10 kg

41. From a circular disc of radius R, a square is cut out as shown in figure. The distance of centre of mass of the remaining section from the centre of original disc, is



(1) $\frac{R}{4\pi}$

(2) $\frac{R}{4\pi+2}$

(3) $\frac{R}{4\pi-2}$

(4) $\frac{R}{3\pi+1}$

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42. A light, inextensible string wrapped around a uniform solid cylindrical pulley of mass M = 6 kg and radius R = 0.2 m. A block of mass m = 2 kg hangs from free end of string. The cylinder is free to rotate about its fixed horizontal axis (no slipping between string and rim of pulley), and the system is released from rest. The linear acceleration of falling block and angular acceleration α of the cylinder is

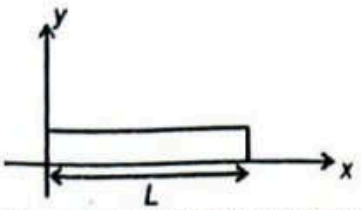
(1) 4 m/s^2, 20 rad/s^2

(2) 20 m/s^2, 4 rad/s^2

(3) 12 m/s^2, 10 rad/s^2

(4) 10 m/s^2, 10 rad/s^2

43.



Consider a rod of length L placed as shown. The position of its centre of mass (x_{CM}) from origin if its linear mass density is given by $\lambda = \frac{\lambda_0 x^2}{L^2}$ (where λ_0 is a constant), is

(1) $\frac{L}{1}$

(2) $\frac{2L}{3}$

(3) $\frac{3L}{4}$

(4) $\frac{L}{2}$

44. A bomb of mass 2 kg is projected with speed 20 m/s at an angle 60° from horizontal. At the topmost point, it exploded into two equal fragments. If one fragment retraces its path, then velocity of other fragment just after the explosion, will be

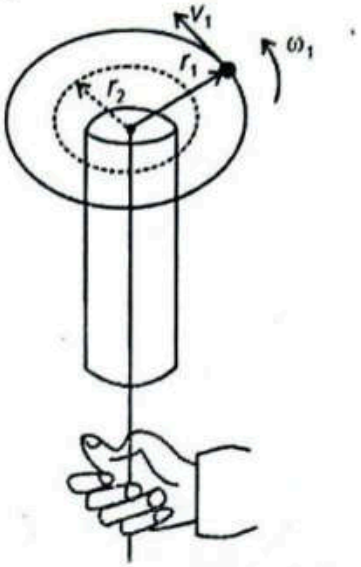
(1) 30 m/s

(2) 40 m/s

(3) 50 m/s

(4) 60 m/s

45. A point mass is tied to one end of a cord whose other end passes through a vertical hollow tube, which is held by a person. The point mass is rotating in a horizontal circle of radius 2 m with a speed of 4 m/s as shown below. The cord is then pulled down so that the radius of circle reduces to 1 m. The new linear and angular velocities of the point mass is



(1) 4 m/s, 8 rad/s

(2) 8 m/s, 8 rad/s

(3) 5 m/s, 4 rad/s

(4) 5 m/s, 5 rad/s

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CHEMISTRY

46. Match List-I with List-II

	List-I (Species)		List-II (Bond angle)
(a)	BF_3	(i)	Contains only 180° and 90° bond angle
(b)	XeF_4	(ii)	Contains only 120° bond angle
(c)	PCl_5	(iii)	Contains 90°, 180° and 120° bond angles
(d)	XeF_2	(iv)	Contains only 180° bond angle

The correct match is

(1) a(ii), b(iii), c(i), d(iv)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(i), b(ii), c(iv), d(iii)

47. If a process is spontaneous at only high temperature, then

(1) $\Delta H < 0$ and $\Delta S > 0$

(2) $\Delta H > 0$ and $\Delta S > 0$

(3) $\Delta H > 0$ and $\Delta S < 0$

(4) $\Delta H < 0$ and $\Delta S < 0$

48. The IUPAC official name of element having atomic number Z = 104 is

(1) Bohrium

(2) Rutherfordium

(3) Meitnerium

(4) Nobelium

49. For the reaction,
$N_2H_2(g) + H_2(g) \rightarrow N_2H_4(g)$, $\Delta_r H^\circ = -109$ kJ / mol

The bond energy of (N – N) bond will be

Given : BE of (N = N) = 400 kJ/mol
BE of (H – H) = 436 kJ/mol
BE of (N – H) = 391 kJ/mol

(1) 163 kJ/mol

(2) 206 kJ/mol

(3) 90 kJ/mol

(4) 54 kJ/mol

50. Given below are two statements
Statement I : Electronegativity of phosphorus is less than that of carbon in Pauling scale.
Statement II : Atomic radii of phosphorus is less than that of boron.
In the light of above statements, choose the correct answer from the options given below.

(1) Both statements I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

51. 1 mole of an ideal gas at 300 K and 10 atm is expanded to 1.0 atm pressure isothermally and reversibly then work involved is

(1) –5.74 kJ

(2) –5744 kJ

(3) –57.44 kJ

(4) –5.7 J

52. The correct relationship between C_P and C_V for 10 moles of an ideal gas is
(C_P and C_V are molar heat capacity at constant pressure and constant volume respectively)

(1) $C_P - C_V = 10R$

(2) $C_P - C_V = \frac{R}{10}$

(3) $C_P - C_V = R$

(4) $C_P + C_V = 10R$

53. Which of the following are correctly matched?

Element	Atomic number	Period Number	Group Number
(A)	40	4	4
(B)	73	7	1
(C)	105	7	5
(D)	53	5	17

Choose the correct option

(1) (C) and (D) only

(2) (A), (C) and (D) only

(3) (B) and (C) only

(4) (A) and (D) only

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54. Which of the following represents the correct order of increasing negative electron gain enthalpy for the elements O, S, Se and Te?

(1) Te < Se < S < O

(2) O < S < Se < Te

(3) O < S < Te < Se

(4) O < Te < Se < S

55. The heat required to raise the temperature of 25 g of copper from 35°C to 80°C is

(specific heat of copper = 0.39 J g^{-1} $°C^{-1}$)

(1) 625.25 J

(2) 438.75 J

(3) 375.31 J

(4) 875.25 J

56. Match List-I with List-II.

List-I Species	List-II Hybridisation and shape
a. SO_2	(i) sp^3d^2 and square planar
b. XeF_4	(ii) sp^3d^2 and square pyramidal
c. $\overset{\oplus}{N}H_4$	(iii) sp^2 and bent
d. BrF_5	(iv) sp^3 and tetrahedral

Choose the correct option.

(1) a(iii), b(ii), c(iv), d(i)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(iii), b(i), c(iv), d(ii)

57. Which of the following does not exist?

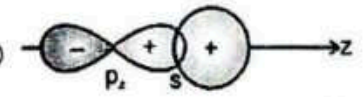
(1) H_2

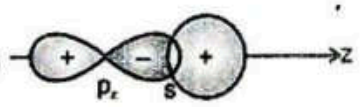
(2) He_2

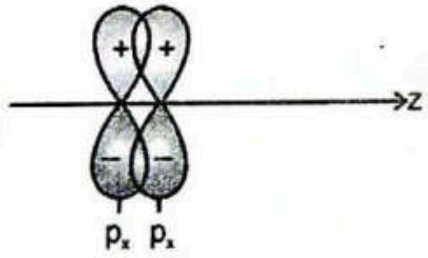
(3) B_2

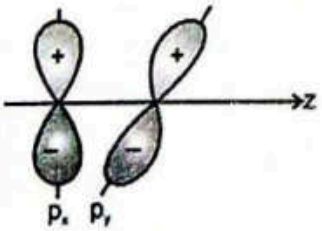
(4) N_2

58. Which among the following will result into negative overlap of atomic orbitals?

(1) 

(2) 

(3) 

(4) 

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59. The correct decreasing order of O–O bond length for the given species is

(1) $O_2^+ > O_2 > O_2^-$

(2) $O_2^- > O_2 > O_2^+$

(3) $O_2 > O_2^- > O_2^+$

(4) $O_2^- > O_2^+ > O_2$

60. Consider the following pairs of species:

(a) Mn^{2+} and Fe^{3+}

(b) Al^{3+} and F^-

(c) Ni^{2+} and Fe^{2+}

(d) O^{2-} and Ne

Which of the given pairs is/are not isoelectronic pairs?

(1) (b) and (c) only

(2) (c) only

(3) (a), (b) and (d) only

(4) (b), (c) and (d) only

61. Given below are the two statements

Statement I : O_3 molecule is a resonance hybrid of three canonical structures.

Statement II : Resonance stabilizes the molecule as the energy of the resonance hybrid is less than the energy of any single canonical structure.

In light of above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

62. If the observed dipole moment of A–B molecule is 1 D and A – B bond length is 100 pm then percentage ionic character in A–B molecule will be

(1) 25.75%

(2) 16.65%

(3) 20.81%

(4) 14.29%

63. Correct order of bond order among the following is

(1) $C_2^{2-} < C_2^- < C_2$

(2) $C_2^- < C_2^{2-} < C_2$

(3) $C_2^- < C_2 < C_2^{2-}$

(4) $C_2 < C_2^- < C_2^{2-}$

64. Consider the following statements
(a) NO and NO_2 are odd electron molecules
(b) Expanded octet is observed in PF_5 and H_2SO_4
(c) Octet rule is applicable in CCl_4 and SCl_2
(d) Shape of molecules can be explained by octet theory
The correct statements are

(1) (c) and (d) only

(2) (a) and (b) only

(3) (a), (b) and (c) only

(4) (a), (b), (c) and (d)

65. The molecule having highest lattice energy among the following is

(1) LiBr

(2) NaCl

(3) LiCl

(4) NaBr

66. Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R)
Assertion (A): Both O_2^- and N_2^- species are paramagnetic.
Reason (R): Both O_2^- and N_2^- species have unpaired electron(s) in π^* molecular orbital.
In the light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

67. Compare the bond dissociation enthalpies of the following species : H_2, H_2^+ and H_2^-

(1) $H_2 < H_2^+ < H_2^-$

(2) $H_2^+ < H_2 < H_2^-$

(3) $H_2^+ < H_2^- < H_2$

(4) $H_2^- < H_2^+ < H_2$

68. In a process, 600 J of heat is absorbed by the system and 250 J of work is done by the system, change is internal energy during the process is

(1) 100 J

(2) 850 J

(3) 350 J

(4) 250 J

69. Correct order of ionisation energy of Li, Be, B and C is

(1) C > Be > B > Li

(2) Be > C > B > Li

(3) C > B > Be > Li

(4) B > Be > C > Li

70. The set which does not represent Dobereiner's Triads is

(1) Zn, Cd, Hg

(2) Li, Na, K

(3) Ca, Sr, Ba

(4) Cl, Br, I

71. Consider the following statements
(a) The energy of an isolated system is constant.
(b) Both heat and work are path functions.
(c) For isothermal expansion of an ideal gas into vacuum, w = 0.
(d) Both q_p and q_v are state functions (q_p and q_v are heat absorbed at constant pressure and constant volume respectively).
The correct statements are

(1) (a) and (d) only

(2) (b), (c) and (d) only

(3) (b) and (c) only

(4) (a), (b), (c) and (d)

72. Match List-I with List-II

	List-I (Molecule)		List-II Dipole moment, μ(D)
(a)	H_2O	(i)	1.04
(b)	NH_3	(ii)	1.85
(c)	$CHCl_3$	(iii)	0.38
(d)	HI	(iv)	1.47

Choose the correct match

(1) a(i), b(iii), c(ii), d(iv)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(ii), b(iii), c(iv), d(i)

73. For the reaction,
$C_6H_5COOH(s) + \frac{15}{2}O_2(g) \rightarrow 7CO_2(g) + 3H_2O(l)$
The difference between enthalpy change and internal energy change at temperature T (K) is

(1) +RT

(2) –RT

(3) $\frac{3}{2}RT$

(4) $-\frac{RT}{2}$

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74. Given below are two statements

Statement I: For both reversible and irreversible expansion for an ideal gas, under isothermal conditions, $\Delta U = 0$ and $\Delta S_{total} = 0$.

Statement II: Standard molar enthalpies of formation ($\Delta_f H^\circ$) at 298 K is zero for both $H_2(g)$ and $Br_2(g)$.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

75. Consider the following statements about hybridisation.

(a) Promotion of electron is an essential condition prior to hybridisation.

(b) The hybridised orbitals are always equivalent in energy and shape.

(c) The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.

The correct statement(s) is/are

(1) (b) and (c) only

(2) (a) only

(3) (a) and (c) only

(4) (b) only

76. For the reaction, $4A(g) + B(s) \rightarrow 4C(g)$, if $\Delta U^\circ = -10$ kJ and $\Delta S^\circ = -40\ J\ K^{-1}$, then value of ΔG° at 298 K will be

(1) –1.92 kJ

(2) 1.92 kJ

(3) 0.55 kJ

(4) –0.55 kJ

77. Consider the following processes

(a) $Na(g) + e^- \rightarrow Na^-(g)$

(b) $Br(g) + e^- \rightarrow Br^-(g)$

(c) $Kr(g) + e^- \rightarrow Kr^-(g)$

(d) $O^-(g) + e^- \rightarrow O^{2-}(g)$

(e) $S(g) + e^- \rightarrow S^-(g)$

The endothermic process(es) is/are

(1) (a), (b) and (e) only

(2) (b) and (c) only

(3) (c) and (d) only

(4) (e) only

78. Consider the following statements

(a) Strontium and antimony are representative elements

(b) Plutonium and americium are transuranium elements

(c) Zn, Cd and Hg are not transition elements

The correct statements(s) is/are

(1) (a) and (c) only

(2) (a), (b) and (c)

(3) (a) only

(4) (b) and (c) only

79. Match List-I with List-II

	List-I (Molecular orbital)		**List-II Nodal plane(s)**
(a)	$\sigma 1s$	(i)	3
(b)	$\sigma^* 2p_z$	(ii)	1
(c)	$\pi^* 2p_x$	(iii)	2
(d)	$\sigma^* 1s$	(iv)	0

The correct match is

(1) a(iv), b(iii), c(i), d(ii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(ii), b(iv), c(i), d(iii)

80. If the equilibrium constant of a reaction is 2×10^3 at 25°C then the standard Gibbs free energy change for the reaction will be nearly

(1) $-2.5R \times 298$

(2) $-7.6R \times 298$

(3) $-7.6R$

(4) $-5.1R \times 298$

81. Consider the following statements

(a) Henry Moseley showed that atomic number is a more fundamental property of an element than its atomic mass

(b) Newland's law of Octaves seemed to be true only for elements up to calcium.

(c) Eka-Silicon and Eka-Aluminium are gallium and germanium respectively.

Choose the correct statements

(1) (a) and (b) only

(2) (a) and (c) only

(3) (b) and (c) only

(4) (a), (b) and (c)

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82. The heat of combustion of C(graphite) to $CO_2(g)$ is $-x$ kJ/mol. The heat released upon formation of 88 g of $CO_2(g)$ from C(graphite) and oxygen gas is

(1) 1.5x kJ

(2) 0.5x kJ

(3) 2x kJ

(4) 3.0x kJ

83. Bond dissociation energy of CH_4 is 360 kJ/mole and C_3H_8 is 820 kJ/mole. The bond dissociation energy of (C – C) bond is

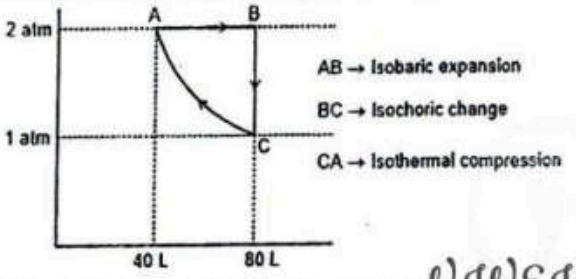
(1) 50 kJ/mole

(2) 80 kJ/mole

(3) 120 kJ/mole

(4) 210 kJ/mole

84. One mole of an ideal gas undergoes following processes



The total work involved in the above process will be nearly

(1) – 24.5 L atm

(2) – 16.4 L atm

(3) – 20.8 L atm

(4) – 31.1 L atm

85. Given below are two statements

Statement I: Total number of lone pairs of electrons in ClF_3 molecule is eleven.

Statement II: ClF_3 molecule has sp^3d hybridisation with 90° bond angles.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

86. Given below are two statements, one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): F has less negative electron gain enthalpy than Cl.

Reason (R): When an electron is alked to F the added electron goes to the smaller n = 2 quantum level and suffers significant repulsion from the other electron present in this level.

In the light of above statements choose the correct answer.

(1) Both (A) and (R) are true and (R) is correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

87. Consider the following statements.

(a) In o-Nitrophenol, intramolecular hydrogen bond is present.

(b) HF has stronger hydrogen bonding than HCl.

(c) Hydrogen bond is stronger than the covalent bond.

(d) The magnitude of hydrogen bonding depends on the physical state of the compound.

The correct statement(s) is/are

(1) (a) only

(2) (b) and (d) only

(3) (a), (b) and (d) only

(4) (a), (b), (c) and (d)

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88. In which of the following processes, entropy increases?

(a) A liquid crystallizes into a solid.

(b) Temperature of a crystalline solid is raised from 0 K to 115 K

(c) $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$

(d) $H_2(g) \rightarrow 2H(g)$

The correct option is

(1) (a) only

(2) (b) and (c) only

(3) (a), (b) and (c) only

(4) (b), (c) and (d) only

89. The correct order of metallic characters of the given elements is

(1) P > Na > Be > Si

(2) Na > Be > Si > P

(3) P > Si > Na > Be

(4) Na > Si > P > Be

90. Consider the following reaction

$BF_3 + NH_3 \rightarrow F_3B{\cdot}NH_3$

The hybridisation of B and N atoms in the product respectively are

(1) sp^2, sp^3

(2) sp^2, sp^2

(3) sp^3, sp^3

(4) sp^3, sp^2

BOTANY

91. Match List-I with List-II.

	List-I		List-II
a.	Number of leaflets are present on the rachis	(i)	Silk cotton
b.	A single leaf arises at each node in alternate manner	(ii)	Neem
c.	More than two leaves arise at a node and form a whorl	(iii)	*Petunia*
d.	The leaflets are attached at a common point	(iv)	*Alstonia*

Choose the **correct** option.

(1) a(ii), b(iii), c(i), d(iv)

(2) a(i), b(iv), c(iii), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(ii), b(iii), c(iv), d(i)

92. Select the **correctly** matched pair.

(1) Monoadelphous – Pea

(2) Diadelphous – China rose

(3) Syncarpous – Mustard

(4) Apocarpous – Tomato

93. Which of the following statements is **correct**?

(1) The position of mother axis with respect to the flower is represented by a dot on the bottom of the floral diagram

(2) The seed coat has two layers, the outer testa and inner tegmen

(3) Adventitious roots arise from different parts of plants as well as from radicle

(4) Stems are the most important vegetative organs for photosynthesis in mesophytes

94. A floral diagram provides information about the arrangement of

(i) Flowers on the floral axis

(ii) Different floral parts within a flower

(iii) Ovules within the ovary

The **correct** ones are

(1) (ii) and (iii) only

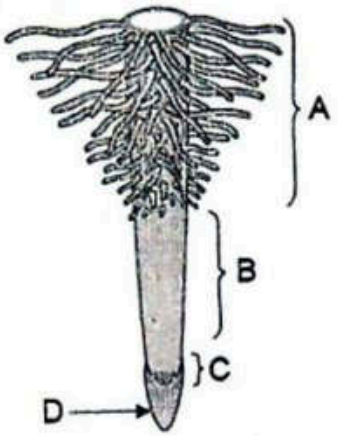
(2) (i) and (iii) only

(3) All (i), (ii) and (iii)

(4) (i) and (ii) only

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95. Examine the following figure. State true (T) or false (F) for the statements (i–v) w.r.t. the labels (A-D) and choose the **correct** option.



(i) Label D protects the tender apex of root as cells are covered with thick layer of suberin and cuticle.
(ii) The cells of the label B are ultimately differentiated and then undergo rapid elongation.
(iii) The cells which are present few millimeters just above the label D are thin-walled, have dense protoplasm and are comparatively larger than other cells due to having larger nuclei.
(iv) Some of the epidermal cells of label A form structures that help in water and mineral absorption from the soil.
(v) The cells of the label C divide repeatedly to form permanent tissues after differentiation.

	(i)	(ii)	(iii)	(iv)	(v)
(1)	T	F	F	T	F
(2)	F	F	F	T	T
(3)	F	T	F	T	F
(4)	F	F	F	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

96. The scutellum observed in a grain of wheat is comparable to which part of the seed in Pea?

(1) Plumule

(2) Cotyledon

(3) Endosperm

(4) Aleurone layer

97. Match the column-I with column-II and select the **correct** option.

	Column-I		Column-II
a.	Contain pigments that are same as those present in higher plants	(i)	They appear yellow, green, brown, blue or red, depending on the main pigment present in their cells
b.	They play a great role in recycling nutrients	(ii)	Majority of them are fresh water organisms found in stagnant water
c.	Under unfavorable conditions, they produce spores	(iii)	They oxidise various inorganic substances and use the released energy for ATP production
d.	Have two flagella; one lies longitudinally and other transversely in a furrow between the wall plates	(iv)	They depend on dead organic matter for food

(1) a(ii), b(iii), c(i), d(iv)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(ii), b(iii), c(iv), d(i)

98. Select the **correct** option w.r.t. the given characteristics in the following table.
(Here '+' indicates present and '–' indicates absent)

	Fungus	Zoo-spores	Conidia	Coenocytic mycelium	Dikaryo-Phase
(A)	Rust fungus of wheat	–	–	–	+
(B)	*Albugo*	+	–	+	+
(C)	*Aspergillus*	–	+	–	+
(D)	*Colletotrichum*	+	+	–	–

(1) (A) and (B) only

(2) (A) and (C) only

(3) (A) and (D) only

(4) (B) and (D) only

99. Morels and truffles are edible and are considered delicacies. They belong to the group

(1) Ascomycetes

(2) Basidiomycetes

(3) Deuteromycetes

(4) Phycomycetes

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100. Read the following Assertion (A) and Reason (R) statements and choose the **correct** option.
Assertion (A): Venus fly trap and *Cuscuta* are placed in the kingdom Plantae.
Reason (R): Venus fly trap and *Cuscuta* are partially heterotrophic plants.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

101. The main criteria for classification, used by R.H. Whittaker include
(i) Cell structure
(ii) Body organisation
(iii) Mode of nutrition
(iv) Phylogenetic relationship
(v) Reproduction
The **correct** ones are.

(1) (i), (ii) and (iii) only

(2) (i), (iii) and (iv) only

(3) All (i), (ii), (iii), (iv) and (v)

(4) (ii), (iv) and (v) only

102. Read the following statements.
Statement A: Kingdom Protista has brought together *Chlamydomonas* with *Amoeba* which were earlier placed in the kingdom Animalia and Plantae, respectively.
Statement B: Bacteria as a group show the most extensive metabolic diversity.
Select the **correct** answer from the following options.

(1) Statement A is correct but statement B is incorrect

(2) Statement A is incorrect but statement B is correct

(3) Both the statements are incorrect

(4) Both the statements are correct

103. The acellular structure that was first recognised by Dmitri Ivanowsky is similar to the structure that was discovered by T.O. Diener, as both of them
(A) Have RNA as the genetic material.
(B) Can pass through bacteria-proof filters.
(C) Can cause infection in a healthy plant.
(D) Have capsomeres arranged in helical or polyhedral geometric forms that protect the nucleic acid.
(E) Are characterised by having an inert crystalline structure, consisting of polymer of peptide molecules.
(F) Have capacity to replicate by self.
The **correct** ones are

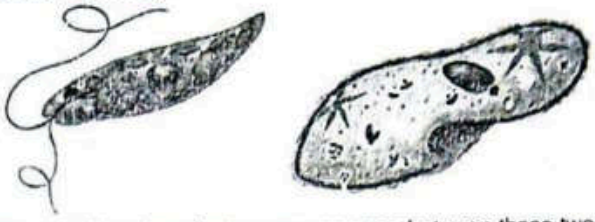
(1) (A), (B) and (C) only

(2) (A), (B) and (D) only

(3) (D), (E) and (F) only

(4) (C), (E) and (F) only

104. Observe the organisms shown in the following figures.



Which of the given features are common between these two organisms?
(a) Single-celled with true nucleus and membrane bound organelles.
(b) Presence of vacuole for osmoregulation.
(c) Reproduce asexually by a process involving cell fusion.
(d) Behave as heterotroph, by predating on other smaller organisms.
(e) Have cilia on the surface and have a gullet that opens to the outside of the cell surface.

(1) (a), (b) and (c) only

(2) (a), (b) and (d) only

(3) (a), (b), (c) and (d) only

(4) (c) and (e) only

105. Mycorrhiza is the symbiotic association between

(1) Algae and fungi

(2) Roots of higher plant and a heterotrophic organism

(3) Photosynthetic part of any plant and a heterotrophic organism

(4) Roots of higher plant and algae

106. How many of the following statements is/are **true** for both *Agaricus* and *Aspergillus* but not for both *Alternaria* and *Mucor*?
(a) Presence of cross walls in their hyphae.
(b) An intervening dikaryotic stage ultimately gives rise to four asexual spores.
(c) Sexual spores are produced exogenously on special mycelium.
(d) Plasmogamy is brought about by the fusion of two vegetative or somatic cells of different sex organs.
(e) After meiosis spores are produced in distinct structures called fruiting bodies.
(f) Zygospore is formed by the fusion of two gametes.

(1) One

(2) Two

(3) Three

(4) Four

107. Prions lack all, **except**

(1) Nucleoprotein

(2) RNA

(3) Pathogenicity

(4) Capsid

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108. One of the classes of organisms that possesses features such as notochord and dorsal hollow neural system is

(1) Chordata

(2) Arthropoda

(3) Mammalia

(4) Insecta

109. Choose the **odd** one out w.r.t. basis of the modern taxonomic studies.

(1) Structure of cell

(2) Evolutionary relationship

(3) Ecological information of organisms

(4) Development process

110. During unfavourable conditions, slime moulds

(1) Act as autotrophs

(2) Form an aggregation

(3) Produce spores

(4) Grow and spread over several feet

111. Given below are some statements.
A - Family has a group of related genera with more number of similarities as compared to genus and species.
B - As we go higher from species to kingdom, the number of common characteristics goes on increasing.
C - Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.
D - Higher the taxa, lesser are the characteristics that the members within the taxon share.
In the light of the above statements, select the **correct** option.

(1) A and B are true and B is the correct reason for A

(2) A and C are true but C is not the correct reason for A

(3) C and D are true and D is the correct reason for C

(4) A and D are true and D is the correct reason for A

112. In the kingdom system of classification proposed by Linnaeus, fungi were placed in the group

(1) Plantae

(2) Animalia

(3) Mycota

(4) Monera

113. Systematics is a branch of study that tells us how different organisms are related to each other. Later its scope became enlarged by including
(a) Identification
(b) Nomenclature
(c) Classification
(d) Evolutionary relationship
The **correct** ones(s) is/are

(1) Only (d)

(2) (a), (b) and (c) only

(3) (a) and (d) only

(4) All (a), (b), (c) and (d)

114. Observe the following expressions w.r.t. the ranks of different taxa in the taxonomical hierarchy.
(a) Arthropoda > Insecta
(b) Diptera > Muscidae
(c) Primata > Hominidae
(d) Sapindales < Poaceae
How many of them is/are represented **correctly**?

(1) One

(2) Two

(3) Three

(4) Four

115. Which among the following scientists was awarded the three prizes, widely regarded as the *triple crown* of biology?

(1) Katherine Esau

(2) Ernst Mayr

(3) Carolus Linnaeus

(4) G. N. Ramachandran

116. *Mangifera indica* and *Triticum aestivum* belong to the same

(1) Class

(2) Division

(3) Order

(4) Family

117. Which of the following statements is **incorrect** w.r.t. the binomial nomenclature?

(1) Biological names are Latinised or derived from Greek, irrespective of their origin

(2) The first word, denoting genus, starts with a capital letter while the specific epithet starts with a small letter

(3) Scientific names are printed in italics and underlined when handwritten

(4) Name of the author is written in abbreviated form after the specific epithet

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118. According to Whittaker's five kingdom classification, how many kingdom(s) include multicellular organisms exhibiting saprophytic mode of nutrition?

(1) One

(2) Two

(3) Three

(4) Four

119. Select the **incorrectly** matched pair.

(1) Prions – Creutzfeldt Jacob disease in humans

(2) Viroids – Potato spindle tuber disease

(3) Bacteriophages – dsDNA acts as the infectious genetic material

(4) Citrus canker – Causal organism has linear DNA

120. Which of the following statements is **incorrect** w.r.t. the members of the kingdom in five kingdom classification whose boundaries are not well defined?

(1) Possess ribosomes that are made up of 60 S and 40 S ribosomal subunits

(2) They are primarily aquatic

(3) They may have flagella for locomotion

(4) Some of them have mechanism to fix atmospheric nitrogen

121. Dikaryotic stage in the life cycle can be seen in

(1) *Trichoderma*

(2) Puffballs

(3) Bread mould

(4) *Colletotrichum*

122. Read the following Assertion and Reason statements and choose the **correct** option.
Assertion : In *Agaricus*, only the vegetative phase is known.
Reason : In *Agaricus*, sex organs are absent.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion

(3) Assertion is false but Reason is true

(4) Both Assertion and Reason are false

123. Margins of petals overlap one another but not in any particular direction in

(1) Lady's finger

(2) China rose

(3) Cotton

(4) Gulmohur

124. Veinlets in the lamina of leaf do not form network. This statement is **true** for

(1) Pea

(2) Wheat

(3) China rose

(4) Mustard

125. In marigold,

(1) Placenta develops at the base of the ovary

(2) Ovules develop on the inner wall of the ovary

(3) Ovary becomes two-chambered due to the formation of false septum

(4) Placenta forms a ridge along the ventral suture of ovary

126. The similar feature between racemose inflorescence and cymose infloresence is that they both have flowers that

(1) Are pentamerous

(2) Are borne laterally in acropetal succession

(3) Are the modified shoots

(4) Are always solitary

127. Which of the following symbols is **not** paired correctly with the description?

(1) % – Zygomorphic flower

(2) $\overset{\frown}{P\ A}$ – Epiphyllous stamen

(3) $\overset{\frown}{C\ A}$ – Epipetalous stamen

(4) $G_{(n)}$ – Apocarpous ovary

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128. The leaf base expands into a sheath, covering the stem partially or wholly in

(1) Gram

(2) Pea

(3) Maize

(4) Tomato

129. In which of the following plants, endosperm is present in mature seeds?

(1) Bean

(2) Orchid

(3) Gram

(4) Wheat

130. _____ is **not** a part of leaf.
Select the **correct** option to fill in the blank.

(1) Stipule

(2) Petiole

(3) Lamina

(4) Pedicel

131. In papilionaceous corolla, the largest petal overlaps the two lateral petals and these lateral petals are known as

(1) Wings

(2) Keel

(3) Standard

(4) Vexillum

132. In maize, embryo is made up of

(1) Radicle, embryonal axis and two cotyledons

(2) Integument, embryonal axis and one cotyledon

(3) Radicle, embryonal axis and one cotyledon

(4) Aleurone layer, radicle and two cotyledons

133. Which of the following characteristics are **true** for both China rose and tobacco?
(i) Axile placentation
(ii) More than two carpels in gynoecium
(iii) Twisted aestivation of corolla
(iv) Superior ovary
(v) Monoadelphous androecium
The **correct** ones are

(1) (i) and (iv) only

(2) (i), (iii) and (iv) only

(3) (iii) and (v) only

(4) (ii) and (iv) only

134. A student was given a twig of a particular plant with a few flowers. He observed the following features.
• Single leaf arise at each node.
• Flower can be divided into two equal radial halves in any radial plane passing through the centre.
• Gynoecium occupies the highest position while the other parts are situated below it.
Based on the above information, how many of the below given plants exhibit(s) all three of these features?
Calotropis, China rose, Guava, Pea, Chilli, Belladonna, Mustard

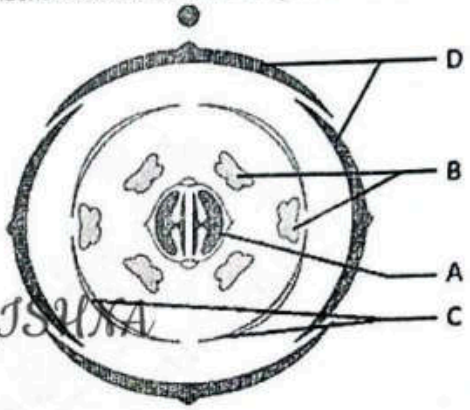
(1) Two

(2) One

(3) Four

(4) Three

135. Figure shows the floral diagram of the member of a family with labels A to D. Select the option which gives correct identification of label and its characteristic.



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(1) B-Stamens – Show variation in the length of filaments within a flower

(2) A-Ovule – Axile placentation

(3) C-Petals – Imbricate aestivation

(4) D-Sepals – valvate aestivation

ZOOLOGY

136. In adult frogs, three-chambered heart is present with two atria and one ventricle. The blood pumped by the ventricle is of mixed type. The left atrium receives oxygenated blood from

(1) Gills and skin

(2) Moist skin only

(3) Lungs only

(4) Skin and lungs

137. Choose the option representing the correct route for air passage during forceful expiration from lung alveoli towards exterior.

(1) Alveoli → Bronchi → Bronchioles → Pharynx → Larynx

(2) Alveoli → Bronchioles → Trachea → Pharynx → Larynx

(3) Alveoli → Bronchioles → Bronchi → Larynx → Pharynx

(4) Alveoli → Bronchi → Larynx → Trachea → Pharynx

138. Select the **incorrect** match w.r.t. humans

(1)	T-P gap in ECG	–	Heart muscles are electrically silent
(2)	Atrio-ventricular septum	–	Thick fibrous tissue
(3)	Dub to Lub	–	Time duration is 0.5 second
(4)	Neutrophils	–	Perform movement by streaming of cytoplasm and have most likely 2-5 nuclei in each

(1) (1)

(2) (2)

(3) (3)

(4) (4)

139. The maximum volume of air a person can breathe in after a forced expiration can be represented by 'X'.
Select the correct representation for 'X'.

(1) [IC + RV] – ERV

(2) [IRV – RV] + TV

(3) [RV + TV] + [ERV – TV]

(4) TLC – RV

140. Consider the given statements and choose the correct option.
Statement (A): If internal urethral sphincter is damaged then it can lead to urine incontinence and can cause infertility in males.
Statement (B): During the counter current interaction between descending limb of Henle's loop and ascending limb of vasa recta, removal of water from the filtrate and return of NaCl to the interstitium helps to maintain the concentration gradient in the medullary interstitium.

(1) Statement (A) is correct but statement (B) is incorrect

(2) Both statements (A) and (B) are incorrect

(3) Both statements (A) and (B) are correct

(4) Statement (A) is incorrect but statement (B) is correct

141. In a blood donation camp, blood collected from a donor with the AB^+ blood group was stored at low temperature (2-6°C) in the blood bank refrigerator. Select the correct option representing most suitable reason w.r.t. it.

(1) Low temperature keeps clotting enzymes active to prevent coagulation.

(2) Low temperature slows down metabolic and enzymatic activities, preventing degradation of blood components and clot formation.

(3) Low temperature causes denaturation of the proteins present in blood plasma.

(4) Low temperature reduces bacterial growth and maximizes haemolysis of RBC, thereby maintaining blood viability.

142. In an enzymatic reaction, an enzyme 'E' catalyzes the conversion of a substrate molecule 'S' into product molecule 'P'. 'I' is introduced to the reaction that act as a competitive inhibitor.
Consider the following statements w.r.t. the above given information and choose the **correct** option.
a. 'I' resembles the substrate in its molecular structure and competes for binding at the enzyme's active site, thereby reducing enzyme activity at low substrate concentrations.
b. At very high substrate concentrations, the inhibition by 'I' can be completely overcome, restoring the normal maximum velocity of the reaction.
c. The presence of 'I' decreases both the V_{max} and binding affinity of enzyme inspite of increasing substrate molecules.
d. For attaining V_{max}, the apparent Km of the enzyme increases in presence of 'I'.
e. 'I' forms a stable enzyme-inhibitor complex that permanently blocks the enzyme's active site, showing irreversible inhibition.

(1) Only statements a, b and d are correct

(2) Only statements a, c and d are correct

(3) Only statements a and e are correct

(4) Only statements c and d are correct

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143. In a hypothetical condition, if a polysaccharide was made using 20 glucose molecules then, how many water molecules will be produced in dehydration synthesis of this compound and what will be its molecular formula respectively?

(1) 20, $C_{120}H_{240}O_{120}$

(2) 19, $C_{120}H_{200}O_{102}$

(3) 19, $C_{120}H_{202}O_{101}$

(4) 20, $C_{120}H_{202}O_{111}$

144. Which among the following molecules will be found in acid-soluble pool obtained upon chemical analysis of a living tissue?

(1) Maltose

(2) Renin

(3) Trypsin

(4) Palmitic acid

145. In the structure of DNA, nucleotides of same strand are attached to each other by __A__, whereas in the structure of a single nucleotide, the sugar moiety is attached to phosphate group through __B__ and a nitrogenous base is joined through __C__ with pentose sugar.
Select the option that **correctly** fills the respective blanks.

	A	B	C
(1)	Phosphoester bond	Phosphodiester bond	Glycosidic bond
(2)	Phosphodiester bond	Phosphoester bond	Glycosidic bond
(3)	Glycosidic bond	Ester bond	Phosphodiester bond
(4)	Ester bond	Glycosidic bond	Phosphodiester bond

(1) (1)

(2) (2)

(3) (3)

(4) (4)

146. Select the correct option to complete the analogy w.r.t. different types of inhibition.
Glucose-6-phosphate and hexokinase : Feedback inhibition :: Malonate and succinic dehydrogenase : _____

(1) Co-factor inhibition

(2) Feedback inhibition

(3) Competitive inhibition

(4) Non-competitive inhibition

147. An individual is suffering from loss of elasticity of walls of bronchioles and damaged alveolar walls due to excessive smoking. Which other conditions will be observed in this individual?
a. Reduced surface area for exchange of gases.
b. Increase in residual volume.
c. Decrease in vital capacity.
Select the option with **correct** conditions only.

(1) a and b only

(2) b and c only

(3) a and c only

(4) a, b and c

148. Which of the following correctly describes the neural regulation of the normal breathing rate and heart rate in a healthy adult human?

(1) Both are primarily regulated by neural regulation through adenohypophysis.

(2) Both are under voluntary control of the forebrain's parietal lobe.

(3) Both are directly controlled by the CNS through higher brain centres present in cerebral cortex.

(4) Both can be modulated by distinct but closely associated centres located in the medulla oblongata of the hindbrain.

149. What will be the effect on respiration rate if the concentration of CO_2 increases in the blood of a human female under certain conditions?

(1) The depth of inspiration decreases and respiration rate increases that causes shallow and fast breathing.

(2) The depth of inspiration and rate of respiration increases, that lengthen the duration of each inspiration.

(3) Depth of breathing and respiration rate both decreases that leads to fast breathing by increasing the duration of inspiration.

(4) The rate of breathing decreases causing increased duration of inspiration that leads to increased depth of respiration along with respiration rate.

150. Select the odd option w.r.t. systemic circulation in humans.

(1) Contraction in the left ventricle pumps blood to capillary beds in organs and tissues throughout the body.

(2) The blood pumped by the left ventricle provides O_2 as well as nutrients to the tissues and takes away CO_2 and waste products.

(3) At the venous end of blood capillaries, CO_2 from cells directly enters into blood and O_2 from blood enters directly into cells without involvement of tissue fluid

(4) The return of deoxygenated blood back to the heart by vena cavae.

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151. How many of the components mentioned in the box below are components of blood plasma?

Fibrinogens, Thrombocytes, Heparin, Haemoglobin, Antibodies, Amino acids, Erythrocytes

Choose the **correct** option.

(1) Seven

(2) Six

(3) Three

(4) Four

152. How does cortical nephrons differ from juxta medullary nephrons?

(1) Former are less specialized for concentrating urine.

(2) Former is responsible for filtration of blood and reabsorption of water and solutes.

(3) Latter do not help in maintenance of fluid and electrolyte balance.

(4) Latter produces hypotonic urine under normal physiological conditions.

153. Choose the correct statement w.r.t. the effect of exercise on the oxyhaemoglobin dissociation curve.

(1) During exercise, the concentration of CO_2 in the muscle increases that leads to shifting of the curve towards the left side.

(2) Temperature in the muscles falls leading to unloading of CO_2 from haemoglobin shifting the curve towards the right side.

(3) The blood becomes alkaline due to fall in pH that shifts the curve towards the left side leading to unloading of oxygen from oxyhaemoglobin to tissues.

(4) Temperature of the muscles increases during exercise that shifts the curve to the right and allowing unloading of oxygen from oxyhaemoglobin to tissues.

154. In a healthy individual under normal physiological conditions, numerical value of haemoglobin present in 100mL of blood is almost equivalent to the percentage of _____ of the total cellular mass.
Choose the most appropriate option to fill in the blank.

(1) Proteins

(2) Carbohydrates

(3) Lipids

(4) Ions

155. Read the following statements.
a. They are small molecular weight compounds than proteins and nucleic acids.
b. They are arranged into structures like cell membranes and other membranes.
c. They are not strictly macromolecules but are found in acid-insoluble pool.
All of the above statements hold true for which of the following compounds?

(1) Glucose

(2) Glycine

(3) Lecithin

(4) Adenosine

156. Select the **incorrect** statement w.r.t. application/importance of proteins.

(1) Proteins like collagen is present as intercellular ground substance

(2) Proteins like insulin is used as therapeutic agent

(3) Protein along with phospholipid is responsible for formation of plasma membrane in animal cells

(4) Protein named glycine is used for the formation of haemoglobin

157. Select the **odd** one w.r.t. polysaccharides.

(1) Chitin

(2) Natural gums

(3) Cellulose

(4) Natural rubber

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158. Haem is an organic co-factor that binds to an enzyme called catalase that catalyzes which of the following reactions?

(1) $2H_2O_2 \rightarrow 2H_2O + O_2$

(2) $C_6H_{12}O_6 + O_2 \rightarrow 2C_3H_4O_3 + 2H_2O$

(3) $CO_2 + H_2O \rightleftharpoons H_2CO_3$

(4) $Ba(OH)_2 + H_2SO_4 \rightarrow BaSO_4 + 2H_2O$

159. If an enzyme is catalysing inter conversion of optical isomers and an another enzyme is catalysing the linking of two compounds, then select the option that correctly represents the classes of above described enzymes respectively.

(1) 6, 5

(2) 5, 6

(3) 4, 5

(4) 5, 4

160. **Statement (A):** In an endothermic reaction, product (P) is at higher energy level than the substrate (S) where one needs to supply energy by heating in order to form the product (P).
Statement (B): Half of the maximum velocity of an enzyme catalysed reaction decreases which indicates presence of a non-competitive inhibitor that binds irreversibly to the active site of the enzyme.
In the light of above statements, choose the **correct** option.

(1) Both statement (A) and statement (B) are correct

(2) Both statement (A) and statement (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

161. 

Above given schematic diagram represents the linear structure of an oligopeptide with the dots representing the amino acids. Select the incorrect statement w.r.t. the above given diagram.

(1) Six water molecules are formed during its dehydration synthesis.

(2) The structure gives positional information of amino acids in this oligopeptide.

(3) The left end is represented by last amino acid and right end is represented by first amino acid.

(4) New amino acid molecule can be added to its C-terminal

162. Select the exclusive property seen in quaternary structure of a protein.

(1) Presence of peptide bonds

(2) Assembly of more than one polypeptide

(3) Structure similar to a revolving staircase

(4) Presence of interchain hydrogen bond

163. Mentioned below are some secondary metabolites. Which among the following belongs to the group of alkaloids?

(1) Codeine

(2) Curcumin

(3) Ricin

(4) Abrin

164. Match the following columns w.r.t. function of the renal tubules.

	Column I		Column II
a.	PCT	(i)	Selective secretion of potassium ions
b.	Descending limb of Henle's loop	(ii)	Small amounts of urea is transported in medullary interstitium
c.	DCT	(iii)	Selective secretion of ammonia
d.	Collecting duct	(iv)	Almost impermeable to electrolytes

Select the **correct** option.

(1) a(iii), b(iv), c(i), d(ii)

(2) a(iv), b(i), c(iii), d(ii)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(iv), b(i), c(ii), d(iii)

165. All of the statements mentioned below hold true for glomerular filtration, **except**

(1) The pores of the glomerular capillaries generally prevent red blood cells from entering the capsular space.

(2) Glomerular capillaries present a large surface area for filtration

(3) The filtration membrane is thin and porous

(4) The afferent arteriole is narrower than the efferent arteriole, connected with glomerulus, creating high pressure to outflow of blood from glomerulus

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166. Select the **correct** option w.r.t. functions of ANF.

(1) Increases blood volume and blood pressure

(2) Increases the glomerular filtration rate

(3) Causes constriction of afferent and efferent glomerular arteriole

(4) Suppresses the release of aldosterone from adrenal medulla

167. Which among the following conditions will not lead to excess urine formation?

(1) ADH levels are low

(2) The body is overhydrated

(3) ANF levels are high

(4) Vasopressin levels are high

168. Although an artificial kidney can efficiently remove nitrogenous waste and excess salts from a patient's blood during haemodialysis, it still cannot act as a complete substitute for the natural kidney because it fails to

(1) Maintain the osmotic concentration of body fluids during the process

(2) Allow the passage of molecules based on concentration gradient

(3) Synthesise a hormone that stimulates erythrocyte production

(4) Filter plasma through semi-permeable membranes

169. In the human kidney, if active NaCl transport from the ascending limb of Henle's loop is selectively inhibited while functions of all other portions of renal tubule remain normal, which of the following effects would most likely occur?

(1) Decreased NaCl concentration in the medullary interstitium, reducing the medullary osmotic gradient

(2) Increased water reabsorption from the descending limb due to enhanced hypertonicity of medullary interstitium

(3) Increased urine concentration because of higher ADH activity

(4) No effect on medullary osmolarity, as urea recycling alone maintains the same gradient

170. The respiratory system in humans helps in

a. Regulating blood pH
b. Olfaction, as it contains receptors for sense of smell
c. Sound production
d. Eliminating some water from the body

Select the **correct** option

(1) a, b and d only

(2) b, c and d only

(3) a, c and d only

(4) a, b, c and d

171. The affinity of haemoglobin for oxygen decreases at the tissue level. Which combination of the following factors is primarily responsible for this change?

(1) ↑pCO_2, ↓temperature, ↑pH

(2) ↓pCO_2, ↑temperature, ↑pH

(3) ↑pCO_2, ↑temperature, ↓pH

(4) ↓pCO_2, ↓temperature, ↓pH

172. Receptors associated with aortic arch and carotid artery can recognise which of the following changes in arterial blood?

(a) ↑CO_2 concentration

(b) ↑H^+ concentration

(c) ↓O_2 concentration

Choose the correct option.

(1) (a), (b) and (c)

(2) (a) and (b) only

(3) (b) and (c) only

(4) (a) and (c) only

173. Identify the **correct** event taking place at the time of normal expiration.

(1) The external inter-costal muscles contract to increase the volume of thoracic cavity in dorso-ventral axis.

(2) The intra-pulmonary pressure increases than the atmospheric pressure.

(3) The pressure inside the lungs is less than the pressure in pleural cavity.

(4) The diaphragm becomes flat in order to decrease the intra-pulmonary pressure.

174. Which of the following is **correct** w.r.t. the events of cardiac cycle?

(1) When semilunar valves are closed blood is ejected into the arteries.

(2) Left and right atria contract, pushing blood into the ventricles of their respective sides.

(3) Ventricles are relaxed when artrioventricular (AV) valves are closed.

(4) Ventricular pressure rises above atrial pressure leading to opening of the AV valves.

175. In a standard ECG, QT interval represents

(1) Ventricular depolarisation and repolarisation

(2) Joint diastole

(3) Only ventricular diastole

(4) Duration of a cardiac cycle

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176. If the smooth muscles of the tunica media of veins are damaged, which of the following is the most likely consequence?

(1) Veins will lose tone, leading to blood pooling in the lower extremities and reduced venous return to the heart.

(2) Veins will constrict excessively causing hypertension

(3) Veins will automatically constrict to compensate for loss of smooth muscle tone.

(4) Veins will still maintain normal venous return without relying on skeletal muscle or valves.

177. Select the **incorrect** match.

(1) Ventricular diastole – AV valves remain open

(2) Atrial systole – AV valves remain closed

(3) Ventricular systole – AV valves get closed

(4) Joint diastole – AV valves remain open

178. Select the **correct** match w.r.t. disorders of heart.

(1)	Atherosclerosis	–	Deposition of fat, cholesterol and calcium which makes the lumen of coronary arteries narrower
(2)	Heart failure	–	Heart stops beating
(3)	Heart attack	–	Heart is not pumping blood effectively but heart muscles are normal
(4)	Cardiac arrest	–	Heart muscles are suddenly damaged

(1) (1)

(2) (2)

(3) (3)

(4) (4)

179. Consider the following statements.

Statement (A): The chordae tendinae are fibrous cord that attaches papillary muscles of the ventricles to AV valves and destruction of these fibrous cords will result in backflow of blood during ventricular systole.

Statement (B): The AV node delays the electrical impulses to allow the atria to completely empty their blood into the ventricles before the ventricular systole.

In the light of above statement, select the **correct** option.

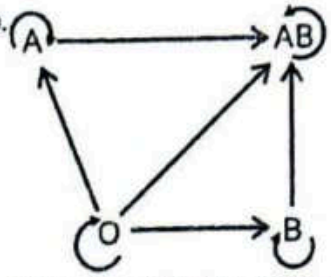
(1) Both statements (A) and (B) are incorrect

(2) Both statements (A) and (B) are correct

(3) Only statement (A) is incorrect

(4) Only statement (B) is incorrect

180.



Read the following statements w.r.t. above given diagrammatic representation of ABO blood grouping.

a. An individual with 'AB' blood group can receive blood from individuals with other blood groups as their blood plasma is devoid of anti-A and anti-B antibodies.

b. An individual with blood group 'A' can receive blood from individuals with blood group A and O.

c. An individual with blood group 'B' can receive blood from an individual whose blood plasma has anti-A antibody.

d. An individual with blood group 'O' can donate blood to individuals that show presence of antigen A or B or both on the surface of RBCs.

How many of the above statements is/are **correct**?

(1) One

(2) Two

(3) Three

(4) Four

VIVEK KRISHNA